

Literature Status Note: Countable Tightness of $P(K)$ and Maharam Type

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Status

Literature already answered, relative-consistency negative. The question extracted from Krupski–Plebanek [1] has a later literature answer under Jensen’s $\diamond$: Koszmider–Silber [2, Theorem 1] construct a compact Hausdorff space K such that $P(K)$ is countably tight in the weak-star topology while K carries a Radon probability measure of uncountable Maharam type.

Source Question

In Section 4 of [1], after observing that tightness $\leq \omega_1$ of $P(K)$ bounds Maharam type by ω_1 , the authors ask:

If $P(K)$ has countable tightness, must every $\mu \in P(K)$ have countable Maharam type?

This is in arXiv:1104.1451, source lines 432–446.

Supporting Answer

Koszmider–Silber [2] explicitly study whether small weak-star topology on spaces of measures can coexist with nonseparable measures. Their Theorem 1 states, assuming $\diamond$, that there is a compact Hausdorff K for which $P(K)$ has countable tightness but K carries a Radon measure of uncountable type. Their abstract states the same conclusion for a Radon probability measure of uncountable Maharam type.

Since a nonzero finite Radon measure may be normalized without changing the Maharam type, this supplies exactly the negation of the implication asked in [1]: in a model of $\diamond$, $P(K)$ is countably tight and some probability measure in $P(K)$ has uncountable Maharam type.

Scope and Limitations

This is not an original result of the run and should not be counted as a new proof or counterexample. It is a retrieval/status result: the exact question is already answered in later literature, modulo the additional set-theoretic axiom $\diamond$.

The answer is not an absolute ZFC construction. Koszmider–Silber note that the extra set-theoretic assumption is necessary in the consistency sense, because Fremlin had shown it is consistent that such compact spaces do not exist. They also state that weakening $\diamond$ to CH is open.

References

- [1] M. Krupski and G. Plebanek, *A dichotomy for the convex spaces of probability measures*, Topology Appl. 158 (2011), 2184–2190. arXiv:1104.1451.
- [2] P. Koszmider and Z. Silber, *Countably tight dual ball with a nonseparable measure*, arXiv:2312.02750.